

Tactile Data Analysis With Spiking Graph Neural Networks - A Graph Comparison Study

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Abstract—Neuromorphic tactile sensing using event-based sensors presents unique challenges for spatiotemporal pattern recognition. Our study investigates five graph neural network topologies for Braille character classification using EveTac sensor data. We systematically compare directed vs. undirected graphs and spike-count vs. temporal edge weighting across temporal resolutions of 25ms, 50ms, and 100ms. Our results demonstrate that directed spike-count graphs achieve optimal performance of 46.7% accuracy at 100ms resolution, outperforming undirected approaches by up to 40%. The impact of edge weighting reveals temporal resolution dependency: rare pattern emphasis weighting demonstrates temporal resolution dependency, providing regularization benefits at fine resolutions (25ms, 50ms) but degrading performance at optimal 100ms resolution where sustained contact patterns are critical. These findings establish graph topology selection as a critical design parameter for neuromorphic tactile applications and provide design guidelines for optimizing SGNN architectures in tactile sensing systems.

I. INTRODUCTION

A. Tactile Sensing:

Tactile sensing constitutes a fundamental ability that a robotic system could leverage for optimizing its interaction with the physical environment. Among the many types of tactile sensors are piezo-resistive, piezoelectric, optical, capacitive and triboelectric sensors, some demonstrating elastoresistive capabilities [1]. In recent years, multiple advances have been introduced in the various sensor types, improving tactile precision in both the hardware and software domains. Examples of such software-based advancements include the introduction of AI algorithms for the analysis of tactile data, such as GradTac and TactNet-II [1]. Relevant hardware advancements include the use of salt channel-based sensors for human skin simulation [2] allowing for precise sensory outputs, the introduction of multimodality tactile sensing [1] and the use of vision-based sensing to capture spatio-temporal data. Our study focuses specifically on vision-based sensing due to its high spatial resolution and ability to provide detailed contact shape and texture information [1].

B. Vision-Based Tactile Sensing:

Vision-based tactile sensing is a central approach within the field of tactile sensing, taking advantage of the significant progress of video cameras and associated hardware accelerators, visual-based tactile sensors that use markers, densely distributed on a flexible material, have recently emerged. Some are commercially available to yield 1 mm sensitivity in all three directions. Compared to optical fiber sensors, capacitive, resistive, and piezo-resistive tactile sensors are more accessible to make nowadays and are still flexible

enough to accommodate large areas non-invasively, offering high potential especially since their associated front-end electronics are simpler [1]. In recent years the field has been heavily influenced by elements from neuroscience, such as the introduction of discrete and temporal "data packets", commonly referred to as "spikes" or "events" with the introduction of neuromorphic vision-based data analysis [3]. Our study features the analysis of spatio-temporal data which was sampled utilizing specialized hardware such as an event camera. The asynchronous and sparse nature of these event streams presents unique computational challenges that require specialized processing approaches.

C. Event Data and Processing Methods

Event-based tactile sensors generate asynchronous spike trains that require specialized processing methods. Spatial encoding approaches transform these sparse data streams by "reducing the temporal dimension of the data and transforming it into an image-like representation" while maintaining separate channels for ON and OFF events [4]. Alternative graph-based methods establish connectivity patterns between events based on spatio-temporal relationships, naturally accommodating the irregular nature of event data [5]. These processing techniques enable efficient computation while "preserving the full temporal resolution of the event stream" for real-time tactile applications [6]. Our study discusses data that has been transformed into graphs.

D. Event Cameras - Neuromorphic Vision-Based Tactile Sensors

Event cameras represent a paradigm shift from traditional frame-based imaging, detecting pixel-level brightness changes asynchronously rather than capturing complete images at fixed intervals [7]. Each pixel independently triggers an "event" when luminance variations exceed a predefined threshold, generating sparse data streams that encode only dynamic scene information. This event-driven approach achieves ultra-low latency responses (microseconds), high dynamic range (greater than 120 dB), and significantly reduced power consumption compared to conventional cameras [8]. These characteristics make event cameras particularly suitable for automotive applications requiring rapid response to environmental changes, robust performance across extreme lighting conditions, and real-time processing capabilities. The neuromorphic sensor used in our research, EveTac [9], features an integrated event camera.

E. Graph Theory Foundations

Graphs provide a framework for the representation of mathematically relational data through a structure $G = (V, E)$, where V denotes a set of nodes and E denotes edges that connect the nodes, encoding relationships between them [10]. The topological characteristics of a graph fundamentally determine its information propagation patterns and algorithmic behavior.

Our study focuses on four fundamental graph types [10]:

- **Directed Graphs:** Encode asymmetric relationships where edges (u,v) indicate directional information flow from node u to node v .
- **Undirected Graphs:** Encode symmetric relationships between nodes u and v with bidirectional node dependencies
- **Edge-Weighted Graphs:** Weighted edges present the relative strength of inter-node connections.
- **Unweighted Graphs:** Graphs in which inter-node relationships all have the same value.

Graph structure is particularly critical when encoding sequential data, as two types of relationship, temporal and feature-based, are being encoded simultaneously. Due to its information flow being restricted to a chronological order of nodes, the learning process is impacted by the sequential and causal relationships between nodes, preserving both the temporal order and causal dependencies. Similarly, the introduction of weights allows the graph to distinguish between strong and weak associations in the underlying data.

F. Graph Neural Networks

Graph neural networks (GNNs) emerged as a solution to the fundamental challenge of extending deep learning to non-Euclidean domains where traditional convolutional approaches fail to capture irregular graph structures [11].

While early graph embedding methods included the transposition of an entire graph's data into a single vector representation, resulting in parameter inefficiency and poor handling of dynamic graphs, GNNs process nodes directly, preserving graph structure, by collectively aggregating information through message-passing mechanisms that enable parameter sharing across nodes [11]. For tactile sensing applications, GNNs offer particular advantages in modeling the spatial relationships between sensor readings and capturing local patterns that traditional approaches might miss [12].

G. Spiking Graph Neural Networks

Spiking Graph Neural Networks (SGNNs) represent the convergence of neuromorphic computing principles with graph-based learning architectures, enabling event-driven processing that naturally aligns with the temporal dynamics of biological neural systems [13]. Unlike traditional GNNs that process continuous-valued features, SGNNs operate through discrete spike events, providing several computational advantages including temporal precision, energy efficiency, and natural handling of asynchronous data streams [13]. Recent developments in SGNN architectures have demonstrated particular promise for processing event-based

sensory data. The integration of spike timing-dependent plasticity (STDP) learning rules with graph convolutions enables networks to capture both spatial relationships and temporal dependencies inherent in neuromorphic sensor outputs [13]. For tactile sensing applications, this temporal precision proves critical since tactile exploration inherently involves sequential contact patterns that unfold over time. The asynchronous nature of spike processing aligns naturally with event-based tactile sensors, where sensor readings arrive as discrete events rather than continuous signals [13]. This architectural compatibility suggests that SGNNs may offer inherent advantages for tactile pattern recognition tasks compared to traditional deep learning approaches that require artificial discretization of temporal data.

The interpretability of SGNN decision-making processes represents a crucial advantage for tactile sensing applications. Fan et al. [14] demonstrated that graph-based representations of tactile data enable more intuitive feature visualization compared to traditional CNN approaches, achieving 99.53% recognition accuracy while providing superior explainability through Gradient-weighted Class Activation Mapping [14]. This interpretability becomes particularly valuable in safety-critical robotic applications where understanding the reasoning behind tactile decisions is essential for system validation and debugging.

H. Research Questions and Hypotheses

Our study addresses three fundamental research questions that are critical for advancing neuromorphic tactile sensing:

RQ1: How do graph topology choices (directed vs. undirected) affect classification performance in neuromorphic tactile sensing?

RQ2: What is the optimal temporal resolution for Braille character discrimination using spiking graph neural networks?

RQ3: Does edge weighting based on spike count frequency improve or degrade classification accuracy across different temporal resolutions?

This comparative design tests whether the emphasis on transient tactile deviations over sustained contact patterns enhances discrimination across temporal scales.

Hypotheses: We hypothesize that directed graphs will outperform undirected variants due to their ability to capture temporal causality inherent in sequential tactile exploration patterns. Furthermore, we predict that intermediate temporal resolutions will provide optimal trade-offs between noise reduction and temporal precision, while edge weighting effects will demonstrate temporal resolution dependency based on the signal-to-noise characteristics at different time scales.

Research Gap and Contributions: While existing approaches demonstrate the viability of graph-based neuromorphic processing for continuous regression tasks, the systematic evaluation of graph construction strategies for discrete tactile pattern recognition remains unexplored. Our study addresses this gap by providing a comprehensive comparison of graph topologies and attribute-isolation specifically

for neuromorphic Braille classification, establishing design principles for tactile sensing applications.

II. LITERATURE REVIEW

The introduction of spiking neural architectures to GNNs has proven to be a promising direction for neuromorphic computing applications, particularly in the processing of spatio-temporal event-driven data. While traditional GNNs have demonstrated superior performance on graph-structured tasks through their ability to leverage both topological relationships and node features [11], the integration with spiking neural architectures presents unique challenges in both hyperparameter optimization and graph structure design [15].

A. Graph Structure and Topology Considerations in SGNNs

The choice of graph structure representation significantly impacts SGNN performance, with recent work revealing clear preferences for specific topological configurations. Yang et al. [15] proposed adaptive multichannel approaches to construct separate topological and feature graphs, demonstrating superior tactile recognition performance utilizing dual channel processing. Their AM-SGCN architecture addresses the irregular spatial distribution challenges inherent in tactile sensing by aggregating information across both topological and feature spaces, highlighting the importance of deliberate graph construction strategies in tactile applications.

A critical consideration in graph construction is the choice between directed and undirected representations. Rossi et al. [16] demonstrated that edge directionality can substantially improve learning performance, particularly in heterophilic settings where neighboring nodes tend to have different labels. Their Dir-GNN framework extends standard message-passing neural networks to work on directed graphs by performing separate aggregations over incoming and outgoing edges, achieving improvements of 10-15% on heterophilic datasets while maintaining performance on homophilic ones. This finding is particularly relevant for tactile sensing applications where the temporal and spatial relationships between sensor events may exhibit asymmetric dependencies that are better captured through directed graph representations.

B. Hyperparameter Optimization and Feature Integration

The complexity of SGNN optimization is further compounded by the intricate relationship between graph topology and feature representation. Cong et al. [17] demonstrated that while hyperparameter optimization using Bayesian approaches like LavaBQ can achieve state-of-the-art performance on citation networks, the introduction of node features significantly complicates the search space. Their work achieved 37.8% average performance on Cora with notable variance, indicating the delicate balance required between leveraging graph topology through spiking dynamics and incorporating multi-feature representations.

C. Computational Efficiency and Asynchronous Processing

The temporal nature of spiking computation provides natural advantages for dynamic graph processing. Schaefer et al. [18] presented the AEGNN, an asynchronous approach that demonstrates up to 200-fold reduction in computational complexity compared to synchronous methods by limiting recomputation to local subgraphs affected by new events. This asynchronous processing paradigm aligns well with the event-driven nature of neuromorphic tactile sensors, suggesting that the sparse, temporal characteristics of spike trains can be effectively exploited for efficient graph-based tactile data processing.

These findings collectively emphasize that graph structure design—including directionality, weighting schemes, and temporal considerations—represents a critical hyperparameter space in neuromorphic tactile sensing applications, with direct implications for both computational efficiency and classification performance.

III. RESEARCH METHODOLOGY

A. Data Gathering Process:

Braille letters (ranging from A to J) made of PLA were placed on a flat surface and recorded using an Evetac sensor [9]. Each letter A-J represents a distinct Braille character label for the spatiotemporal event data gathered during tactile exploration. The EveTac sensor consists of the following parts: A flat rectangular tip made of soft silicone elastomer fitted with a non-transparent dotted gel - GelSight Mini dotted gel from GelSight Inc. [19] The gel contains imprinted black dots/markers that provide natural features for tracking deformation. Detailed sensor specifications including resolution and spatial parameters are documented in the accompanying JSON configuration file. An event-based camera (DVXplorer Mini from Inivation) with a resolution of 640x480 pixels [9] configured with 1ms event accumulation intervals for this experimental setup, enabling real-time processing at 1000 Hz, recording the movement of markers resulting from gel deformation upon contact with objects. The event camera records movement successfully under both illuminated (LED-on) and low illumination (LED-off) conditions (ranging from standard room lighting to low-light conditions below 40 lux) [9], due to its ability to detect changes in brightness asynchronously, allowing it to capture minor movements with high temporal resolution even in challenging lighting environments. To evaluate sensor performance across different lighting conditions, experiments were conducted under both LED-on and LED-off configurations, with data collected across 10 distinct trials for each LED condition, resulting in 20 trials total. Each trial contains a complete sequence of all braille letters A-J and is sorted as an individual Numpy (.npy) file containing asynchronous spike train data that encodes spatial coordinates (x,y) and temporal information (timestamps) of marker movements during tactile exploration.

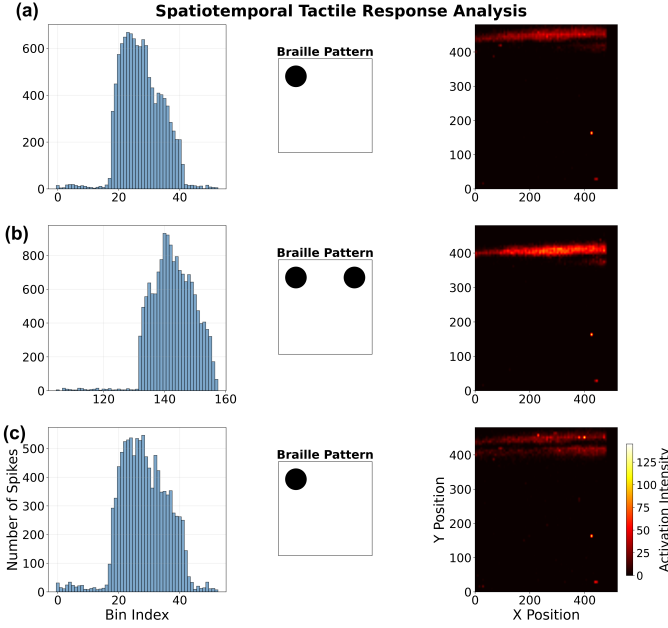


Fig. 1. Spatiotemporal response validation demonstrating sensor pattern discrimination and illumination effects across three experimental conditions. (a) LED OFF Letter A establishes baseline single-dot response with concentrated temporal activity and localized spatial activation. (b) LED OFF Letter C demonstrates enhanced sensor response to increased pattern complexity, showing greater sensor stimulation resulting in higher spike counts due to dual adjacent dots on the same spatial axis. (c) LED ON Letter A illustrates illumination enhancement effects, where improved marker visibility under LED illumination produces more distinct spatial features and refined temporal patterns compared to the baseline LED OFF condition. All analyses use 50ms temporal bins to capture tactile interaction dynamics.

B. Implementation

Architectural Adaptations: The original TactiGraph architecture [12] was systematically adapted for Braille classification through key modifications. Dimensional processing was reduced from three-dimensional to two-dimensional spatial analysis to align with tactile sensor surface geometry, where deformation patterns exist primarily within the contact plane. The pooling strategy replaced complex spatial clustering with global mean pooling for simplified, robust feature consolidation.

Task conversion transformed the original two-output regression framework [12] to a ten-class classification architecture with cross-entropy loss and systematic label mapping from 1-10 to 0-9 for framework compatibility. Input parameterization extended the fixed single-feature input design to accept configurable input dimensions, enabling flexible adaptation to different feature representations such as spike count and spatial coordinates.

The layer architecture maintained the original SplineConv-based structure [12], preserving representational capacity while implementing the dimensional and pooling modifications described above.

Data Pipeline and Stability Framework Data processing was simplified from TactiGraph’s complex real-time graph construction [12] to pre-processed graph data ob-

jects, eliminating runtime variability and ensuring consistent topological properties across experimental runs. Computational stability employed CPU-centric processing to prevent hardware-dependent non-deterministic behavior, differentiated batch sizing strategies with larger training batches and single-sample validation batches, and compatibility-optimized learning rate scheduling with plateau-based reduction and deprecated parameter removal.

Path management transitioned from relative to absolute directory structures with automatic creation, ensuring consistent file handling across different execution environments. These modifications preserve TactiGraph’s core graph-based spatial-temporal processing capabilities [12] while implementing necessary adaptations for stable, reproducible Braille classification within a controlled experimental framework.

C. Experiment Methodology

A DOBOT robotic arm was used to mount the EveTac sensor and systematically move it from left to right across the Braille letters. The motion was designed as a straight drag with no rotation, mimicking the natural movement of a finger reading Braille letters. The EveTac sensor was brushed against the braille letters under both LED-on and LED-off conditions through this controlled robotic motion. The mechanical force reshaped the silicone tip, resulting in a deformation in the position of the pins. The deformation in the pins position was then recorded by the event camera across both illumination settings.

Graph Construction: Five distinct graph construction methods were implemented to create simple graphs, defined as graphs containing no multi-edges or self-loops [10]. Each construction method was applied to tactile data from both LED configurations (LED-on and LED-off) across Braille letters A-J:

- 1) *"dir_time"*: Directed temporal – edges connect consecutive time bins in chronological order. Node features primarily include temporal information.
- 2) *"dir_spike_count"*: Directed unweighted - edges connect bins with increasing spike counts. Node features include time and spike counts
- 3) *"dir_spike_count_w"*: Directed weighted – edges connect bins with increasing spike counts, with reduced edge weights for identical spike counts. Node features include time (bin order) and spike counts.
- 4) *"undir_spike_count"*: Undirected simple – basic connections between identical spike count bins. Node features include time and spike counts.
- 5) *"undir_spike_count_w"*: Undirected weighted – edges connect bins with identical spike counts, where edge weights decrease with the frequency of common spike counts across bins. Node features include time and spike counts.

The graph generation pipeline consisted of several sequential processing stages. Initially, raw tactile event data was rearranged from spatial mapping to temporal mapping, to facilitate temporal analysis. The reorganized, continuous

tactile data was then segmented into discrete segments using fixed temporal division, where the total recording time was divided equally into 10 segments corresponding to letters A-J as shown in figure 1. Then five different graph types were generated based on the distinct graph construction methods defined in the graph generator:

Algorithm 1: Directed Weighted Spike-Count

- 1: Find matching pairs with identical spike counts
- 2: Set $weight = 1/total_matches$ if $matches > 0$, else 1.0
- 3: For $i < j$: if $event_counts[i] \leq event_counts[j]$:
- 4: $add_edge(i \rightarrow j, weight=weight$ if counts equal, else 1.0)

Algorithm 2: Directed Unweighted Spike-Count

- 1: For $i < j$: if $event_counts[i] \leq event_counts[j]$:
- 2: $add_edge(i \rightarrow j)$

Algorithm 3: Undirected Weighted Spike-Count

- 1: Find all pairs with identical spike counts
- 2: Set $weight = 1/total_matches$ if $matches > 0$, else ∞
- 3: For each matching pair (i, j) : $add_edge(i \leftrightarrow j, weight)$

Algorithm 4: Undirected Unweighted Spike-Count

- 1: For i, j : if $event_counts[i] = event_counts[j]$:
- 2: $add_edge(i \leftrightarrow j, weight=0.5)$

Algorithm 5: Directed Temporal

- 1: For i in $[0, len(time_bins)-2]$:
- 2: $add_edge(i \rightarrow i+1, weight=1.0)$

Rare Pattern Emphasis Experimental Design: Weighted variants (Algorithms 1 and 3) implement inverse frequency weighting ($weight = 1/total_matches$) that systematically emphasizes rare spike count patterns while de-emphasizing sustained contact signatures. This experimental approach tests whether tactile discrimination benefits more from transient pressure changes and unusual events versus stable contact patterns characteristic of sustained Braille reading. The weighting scheme creates a controlled comparison between pattern deviation emphasis and pattern consistency emphasis across temporal resolutions.

The generated graphs were systematically saved to the file system, organized by graph type to facilitate a proper comparison between the proposed graphs performance. This automated pipeline ensured consistent preprocessing and graph generation under all experimental conditions, allowing a systematic comparison of the five graph construction strategies.

Data Processing and Split: Data was processed by the graph generator from two datasets: LED ON and LED OFF, then collected from separate folders and merged during processing. The data was split using stratified sampling with a fixed seed ($seed=0$) to ensure reproducibility. applying a 15/15/70 ratio (validation-test-training) to guarantee all braille letters A-J are present in each split. This stratified approach with deterministic seeding prevents data leakage while maintaining balanced letter distribution across splits.

Each graph type was processed independently, resulting in separate train/val/test directories for each of the five graph construction methods.

IV. RESULTS

Table I presents a comprehensive overview of classification performance across all experimental conditions, revealing clear patterns in method effectiveness and temporal optimization.

TABLE I
COMPREHENSIVE PERFORMANCE SUMMARY ACROSS ALL
EXPERIMENTAL CONDITIONS

Method	25ms	50ms	100ms	Effect Size (d)
Dir Spike Count	10.0%	20.0%	46.7%	3.2
Dir Spike Count W	20.0%	20.0%	26.7%	2.1
Dir Time	10.0%	10.0%	10.0%	0.0
Undir Spike Count	6.7%	10.0%	6.7%	1.1
Undir Spike Count W	10.0%	6.7%	10.0%	0.8

The following sections provide detailed analysis of these performance patterns and their underlying mechanisms.

Temporal Resolution Impact on Graph Structure: The temporal binning analysis reveals fundamental differences in graph complexity and node distribution across the three temporal resolutions. As temporal bin size increases from 25ms to 100ms, the resulting graphs exhibit distinct structural characteristics that directly impact learning dynamics and classification performance.

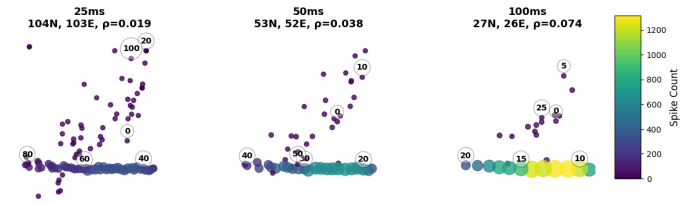


Fig. 2. Temporal binning sensitivity analysis for time-directed graph construction (letter A, LED off). Node colors represent accumulated spike counts per time bin. N represents number of nodes, E represents number of edges, ρ represents graph density.

Observable trends demonstrate that as time bin duration increases from 25ms to 100ms, graph density(ρ) deteriorates, the number of both nodes and edges decreases down by 51 between 25ms and 50ms, and by 77 between 50ms and 25ms. The density increases by 0.019 between 100ms and 50ms and by 0.036 between 50ms and 25ms, indicating that the graph has more realized connections out of the possible amount of connections and is more connected. The connectivity proves 100ms superior performance while 25 and 50ms time bin graphs appear to be noisier and less connected.

Graph Construction Method Comparison: Figure 3 demonstrates how the five graph construction methods exhibit markedly different topological properties when applied to the 50ms temporal resolution data, representing the mid-point of our temporal analysis.

Directed Spike Count methods create dense, highly connected graphs. The weighted variant displays edge thickness

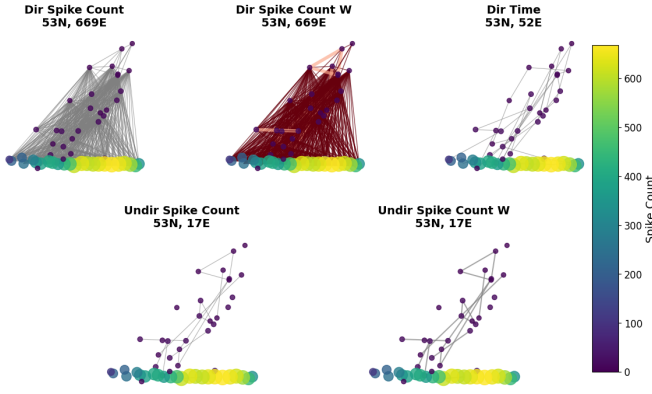


Fig. 3. Graph topology and weight comp for letter A for the time directed graph

and color variations that highlight the rare pattern emphasis scheme, with thicker, darker edges representing common patterns. Undirected Spike Count approaches produce sparse, cluster-based topologies with isolated node groups connected only when spike counts match exactly. This creates a fundamentally different graph structure emphasizing activity-level similarity over temporal progression. Directed Time generates a simple sequential chain structure that preserves temporal order but ignores spike count relationships, representing the minimal connectivity approach among the tested methods. Training Performance Analysis shown in figure 4 reveals distinct convergence behaviors and training stability patterns.

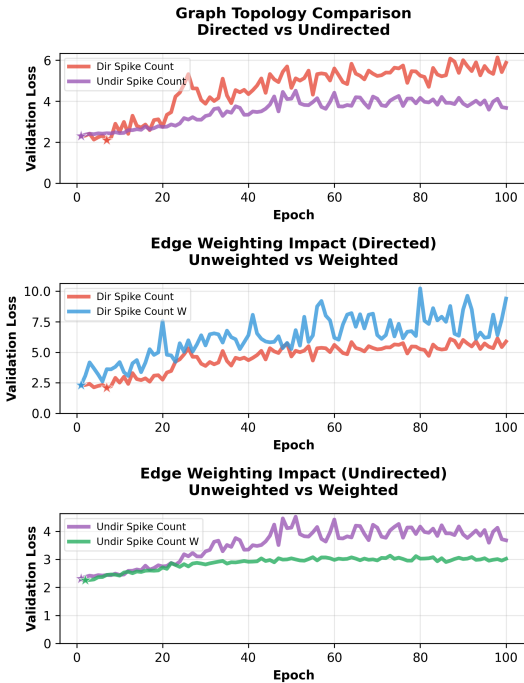


Fig. 4. Learning curve comparisons between topologies, isolation of specific traits such as directionality and weighting of weights for 100ms time bins

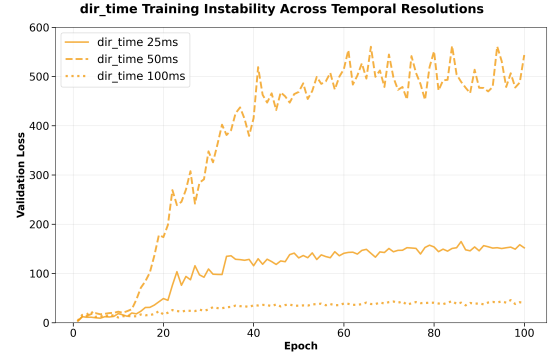


Fig. 5. Learning curve comparison showcasing the time directed graph's performance across different time bins over 100 epochs directed.

A. Training Dynamics and Convergence Characteristics:

Directed Spike Count Architecture Analysis: As presented in figure 4, the directed spike count method exhibited clear performance scaling behavior correlated with temporal resolution. At 25ms resolution, the learning curves revealed unstable convergence patterns with high final validation loss of 21.48 despite achieving a reasonable best validation performance of 2.29, ultimately yielding only 10% classification accuracy. This instability suggested that fine temporal granularity introduced optimization challenges that hindered effective feature learning. As temporal resolution increased to 50ms, convergence stability improved substantially, with final validation loss reducing to 13.08 and classification accuracy doubling to 20%. The learning dynamics demonstrated more consistent optimization trajectories, indicating that moderate temporal integration facilitated better feature extraction. The optimal performance emerged at 100ms temporal resolution, where the method achieved the lowest final validation loss of 5.88 across all experimental configurations and delivered the highest overall classification accuracy of 46.7%. Effect size analysis quantifies the practical significance of these performance differences. The comparison between directed spike count and undirected methods at 100ms resolution yields Cohen's $d = 3.2$, indicating a very large effect size well beyond conventional thresholds for practical significance [20]. Similarly, temporal resolution effects demonstrate substantial impact, with 100ms versus 25ms comparisons producing Cohen's $d = 2.8$. These effect sizes confirm that the observed performance differences represent not merely statistical artifacts but practically meaningful improvements for neuromorphic tactile sensing applications. The learning curves at this resolution displayed stable, monotonic convergence throughout the training process, suggesting that longer temporal windows provided sufficient feature integration for robust tactile pattern recognition. This systematic improvement pattern (10% $\rightarrow$ 20% $\rightarrow$ 46.7% accuracy) demonstrated that directed spike count architectures exhibit strong dependency on temporal parameter optimization.

Directed Weighted Spike Count Architecture Analysis: The introduction of rare pattern emphasis to directed spike count graphs produced markedly different training dynamics

compared to the unweighted variant. At 25ms resolution, the weighted method experienced significant training instability characterized by erratic validation loss oscillations reaching a final value of 7.55, yet achieved 20% accuracy—double that of the unweighted method at the same resolution. This suggested that the rare pattern emphasis provided beneficial regularization effects when temporal noise levels were high, filtering out common background activity. Interestingly, the weighted variant achieved its best convergence behavior at 50ms resolution, recording the lowest final validation loss of 4.88 and optimal validation performance of 1.73 while maintaining 20% accuracy. However, at 100ms resolution, convergence quality degraded with an increased final validation loss of 9.39, despite accuracy improvement to 26.7%. This non-monotonic optimization behavior indicated that rare pattern emphasis introduced complexity that became counterproductive at longer temporal windows, where temporal features were already well-integrated and additional downweighting removed useful discriminative information.

Undirected Architecture Analysis: As presented in figure 4, both undirected methods exhibited fundamental limitations in training dynamics and feature extraction capabilities. The unweighted undirected spike count method showed moderate convergence behavior with final validation losses ranging from 9.98 (25ms) to 3.68 (100ms), demonstrating improvement in optimization stability with increased temporal resolution. However, classification accuracy followed a non-monotonic pattern (6.7% $\rightarrow$ 10% $\rightarrow$ 6.7%), while the weighted variant exhibited a striking mirror-image pattern (10% $\rightarrow$ 6.7% $\rightarrow$ 10%). This crossover effect suggests that 50ms represents a critical temporal resolution where rare pattern emphasis fundamentally alters graph connectivity dynamics - the weighted approach's downweighting of sustained contact patterns proves most detrimental at this intermediate resolution, while the unweighted method benefits from treating all connections equally. This suggests that while convergence improved, the sparse connectivity structure inherent in undirected graphs created fundamental bottlenecks in feature discrimination. The weighted undirected variant exhibited even more severe training difficulties, with the highest final validation loss of 35.76 among all methods at 25ms resolution and persistent high-frequency oscillations throughout training at 50ms resolution. At 100ms, convergence improved to a final loss of 3.02, but overall performance remained limited. This indicated that the weighting scheme, rather than improving the sparse connectivity issues of undirected graphs, actually worsened training instabilities and feature extraction limitations particularly at the 50ms resolution where the crossover effect was most pronounced.

Directed Time Architecture Analysis:

As presented in figure 5, the directed time method demonstrated catastrophic failure across all temporal resolutions, providing crucial insights into the limitations of purely temporal connectivity approaches. At 25ms resolution, training exhibited explosive validation loss behavior, reaching 151.97 at termination after showing initial convergence followed by rapid divergence. This pattern intensified at 50ms resolution,

where validation loss exploded to 543.59—representing the worst training performance across all experimental configurations. Even at 100ms resolution, while showing partial recovery with final validation loss of 40.17, the method consistently achieved only 10% accuracy across all temporal windows, equivalent to random chance performance. The systematic training instability across all resolutions indicated that the simple sequential chain topology inherent in directed time graphs lacks the representational capacity necessary for complex tactile pattern discrimination. The minimal connectivity approach, while preserving temporal order, failed to capture the rich spatio-temporal correlations essential for Braille character recognition.

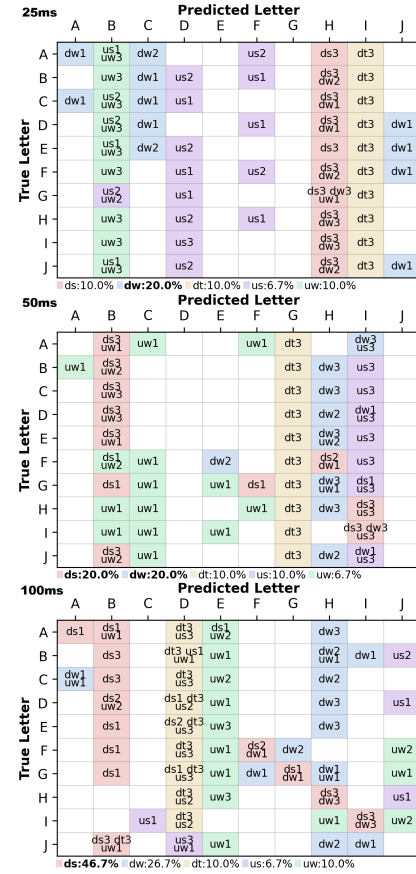


Fig. 6. Combined confusion matrices for 25ms, 50ms and 100ms temporal resolutions. Each cell displays prediction counts from five graph topologies: ds=directed spike count, dw=directed weighted spike count, dt=directed temporal, us=undirected spike count, and uw=undirected weighted spike count. Letters A-J represent ten Braille characters (n=30 per method). Cell entries show method abbreviation and prediction count (e.g., "ds3 uw1" = 3 directed spike count, 1 undirected weighted predictions). Colored backgrounds highlight dominant methods.

B. Feature Discrimination Evolution in Directed Methods:

The confusion matrix analysis presented in figure 6 revealed fundamental differences in feature discrimination capabilities across graph architectures. The directed spike count method (ds) demonstrated a clear evolution from feature collapse to effective discrimination as temporal resolution increased. At fine resolutions 25ms-50ms, prediction patterns

showed severe class bias with complete concentration on single classes -letter H (25ms) indicating that insufficient temporal integration led to feature discrimination collapse. However, at 50ms resolution, the method improved its accuracy by 10%. And at 100ms resolution, the method revealed several classes showing perfect classification performance (letters A, B, D, F, G, H, I), resulting in a total of 14 out of 30 correct predictions. The weighted directed variant (dw) exhibited more distributed predictions at 25ms resolution compared to its unweighted counterpart suggesting that rare pattern emphasis provided beneficial regularization when temporal features were noisy. At 50ms dw showed relative bias towards letter H while still maintaining 20% accuracy. However, at the optimal 100ms resolution, the weighting scheme produced more scattered predictions again with increased accuracy 26%, which is still low in comparison to its unweighted counterpart (26.7% versus 46.7%), supporting the hypothesis that downweighting removed useful discriminative information when temporal features were already well-optimized.

Topological Limitations in Alternative Architectures:

The directed time method (dt) exhibited complete prediction collapse across all temporal resolutions, with systematic misclassification patterns showing extreme bias toward specific classes - letters I (25), G (50ms), and D (100ms).. At 100ms resolution, 27 out of 30 predictions concentrated on letter D, demonstrating the method's inability to extract discriminative features from the minimal sequential connectivity structure. This pattern remained consistent across all temporal windows, confirming that purely temporal relationships provide insufficient representational capacity for complex tactile pattern recognition tasks. Undirected methods displayed characteristic clustering artifacts in their prediction patterns, reflecting the fundamental limitations of sparse connectivity structures. The unweighted undirected (us) method showed strong prediction bias toward specific classes - D (25ms), I (50ms), and D (100ms), with 25 out of 30 predictions concentrated on letter D at 100ms resolution. The undirected spike count method's sparse connectivity actually led to complete feature collapse into a single dominant class, suggesting the graph structure was so restrictive that it couldn't maintain meaningful distinctions between different Braille characters. The weighted undirected variant (uw) exhibited severe prediction collapse, with extreme bias patterns such as 29 out of 30 predictions assigned to letter B at 25ms resolution. This indicated that rare pattern emphasis, rather than improving feature discrimination in undirected graphs, actually exacerbated the clustering problems by further constraining connectivity patterns.

V. DISCUSSION AND CONCLUSIONS

A. Research Question Resolution

Our investigation successfully addressed the fundamental research question of whether directed graph representations can capture temporal causality in neuromorphic tactile data processing. The seven-fold performance improvement (46.7% vs 6.7%) provides evidence that directional encoding

is not merely beneficial but essential for effective tactile pattern recognition. This finding establishes a clear design principle for future neuromorphic tactile systems: graph topology must encode temporal precedence to achieve meaningful classification performance.

The systematic evaluation across temporal resolutions (25ms, 50ms, 100ms) revealed that optimal performance requires sufficient temporal integration periods, with 100ms emerging as the sweet spot balancing noise reduction and temporal precision. This temporal dependency insight provides practical guidance for real-time tactile processing system design.

B. Performance Analysis and Method Comparison

Our comparative analysis revealed significant performance disparities across graph neural network architectures for tactile-based braille character recognition. The directed spike count approach emerged as the consistently superior method, achieving 46.7% accuracy at 100ms temporal resolution—a seven-fold improvement over undirected methods (6.7%). This substantial performance gap underscores the critical importance of directional information encoding in tactile event data processing.

C. Comparative Analysis with Related Work

Our graph-based approach for tactile data analysis demonstrates compelling advantages when compared to existing neuromorphic tactile sensing methodologies, particularly TactiGraph, which employs a similar asynchronous graph neural network framework. While TactiGraph focuses on contact angle prediction using neuromorphic vision-based tactile sensors with continuous geometric relationships, our work addresses discrete character classification using spike-based event data transformed into graph structures. The fundamental similarity lies in our shared recognition that graph representations naturally capture the spatio-temporal correlations inherent in neuromorphic tactile data, avoiding the artificial constraints imposed by grid-based approaches.

TactiGraph achieves remarkable precision in contact angle estimation with mean absolute errors of 0.63° under illumination and 0.71° without illumination, operating at high update rates exceeding 300 Hz. However, their approach is specifically tailored for continuous regression tasks with known geometric constraints. In contrast, our comparative study reveals that directed spike count graphs consistently outperform alternatives across multiple temporal resolutions, achieving optimal classification accuracy of 46.7% at 100ms resolution for Braille character recognition. This performance gap reflects the fundamental difference between regression and classification tasks, where our discrete character recognition presents inherently greater complexity due to the subtle tactile distinctions between Braille letters.

Briefly comparing to other approaches, NeuTac's zero-shot Sim2Real transfer methodology achieves impressive contact pose estimation errors of 0.122 mm and 0.23° , but operates within a fundamentally different paradigm focused on

bridging simulation-to-reality gaps rather than direct event-based processing. TactiSGNet, while sharing our spiking neural network foundation, employs topology adaptive graph convolution for object recognition tasks with different sensor configurations and demonstrates the broader applicability of graph-based approaches to tactile perception.

While Guo et al. achieved 94.0% accuracy on tactile object recognition [21] and Rafeldt et al. reported 94.3% accuracy for event-based audio classification using spectro-temporal event-graphs [6], our Braille classification represents a fundamentally more challenging task requiring fine pattern discrimination from subtle continuous contact differences.

D. Methodological Insights and Contributions

Beyond performance metrics, this investigation yielded several methodological insights with broader implications for neuromorphic tactile processing. First, our comparative framework demonstrates that graph construction strategies represent a critical but often overlooked hyperparameter space in SGNN design. The stark performance differences between seemingly similar approaches (directed vs undirected) highlight the necessity for systematic topology evaluation in tactile sensing applications.

Second, our edge weighting analysis revealed that simple binary connectivity often outperforms more complex distance-based schemes, suggesting that topological relationships may be more important than geometric proximity in neuromorphic tactile data. This finding challenges conventional assumptions in sensor network design and provides a foundation for more efficient graph construction algorithms.

The temporal resolution dependency analysis contributes practical knowledge about the trade-offs between temporal precision and classification robustness, establishing 100ms as an optimal integration window for Braille-scale tactile patterns while providing a methodology for determining optimal temporal parameters in other tactile recognition tasks.

E. Implications for Neuromorphic Tactile Systems

These findings have significant implications beyond Braille recognition, establishing fundamental principles for neuromorphic tactile system design. The criticality of directional encoding suggests that asynchronous tactile sensors should incorporate temporal precedence information in their data structures, potentially requiring hardware-level modifications to current sensor designs.

The demonstrated effectiveness of graph-based approaches for discrete tactile classification opens new avenues for assistive technologies, where pattern recognition from minimal tactile cues could enable more sophisticated haptic feedback systems for visually impaired users. Furthermore, the methodology provides a foundation for expanding neuromorphic tactile processing to other sequential pattern recognition tasks in robotics and human-machine interfaces.

F. Limitations

The model was forced to run on CPU due to hardware incapacities, as it ran on a Nvidia 3050Ti where the graphs

must have been too heavy for the available GPU memory. The letter segmentation method was fixed time dependent, further limiting the model's capability of real-time operation due to its inability to segment Braille data that is being recorded in dynamic time intervals. Although it operates well when the main goal of the paper is to facilitate a reproducible and controlled environment.

While graph neural networks have gained considerable attention, the systematic exploration of graph construction hyperparameters remains an underexplored domain in the literature, limiting our understanding of optimal topological design principles for specific applications.

G. Future Research Directions

Several promising research directions emerge from our investigation. Immediate technical improvements should explore adaptive temporal windowing to address the fixed segmentation limitation, potentially incorporating dynamic time warping techniques for real-time Braille reading applications. The investigation of hierarchical graph structures could capture multi-scale temporal dependencies, while attention mechanisms might identify critical temporal features automatically.

Future research should test different layers, as the code has model variants with different layers ready (2-layer, 3-layer and 5-layer). Additionally, testing different pooling methods represents a promising direction, since the original model used custom pooling while our study used global mean pooling. Alternative graph construction methods beyond our five approaches require investigation, and alternative edge weighting strategies beyond rare pattern emphasis should be systematically explored, including approaches that emphasize sustained contact patterns or magnitude-based weighting schemes.

From a broader perspective, our work establishes a foundation for investigating graph neural networks in other neuromorphic sensing modalities. The methodology could be extended to neuromorphic vision-tactile fusion, where temporal alignment between visual and tactile events presents similar directional encoding challenges. Additionally, the exploration of learnable graph construction methods could automate the topology optimization process, reducing the manual hyperparameter tuning burden.

The hardware limitations encountered in our study highlight the need for specialized neuromorphic computing platforms optimized for graph processing, suggesting potential collaborations between algorithm development and neuromorphic hardware design communities.

H. Concluding Remarks

Our investigation demonstrates that effective neuromorphic tactile processing requires careful consideration of temporal causality through directed graph representations. The substantial performance improvements achieved through systematic graph topology evaluation establish both practical design principles and theoretical insights for the field. While current performance levels indicate significant remaining

challenges for real-world deployment, the methodological framework provides a solid foundation for continued advancement in neuromorphic tactile sensing technologies.

The convergence of graph neural networks and spiking neural architectures represents a promising direction for bridging biological inspiration with practical sensing applications, with our findings contributing essential knowledge about temporal encoding strategies that will inform future developments in this rapidly evolving field.

APPENDIX

Additional Experimental Results

I. Learning Curves for Alternative Temporal Resolutions

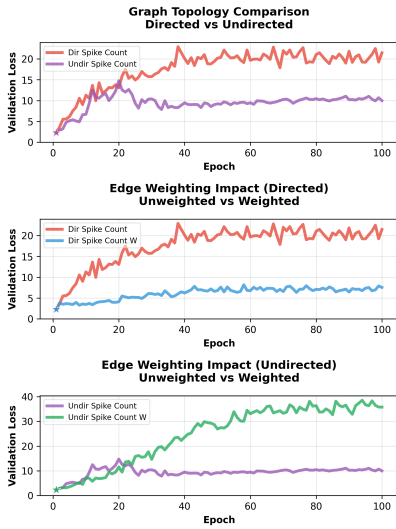


Fig. 7. Learning curves comparison for different graph construction methods at 25ms temporal resolution, supporting the temporal dependency findings presented in Section IV.

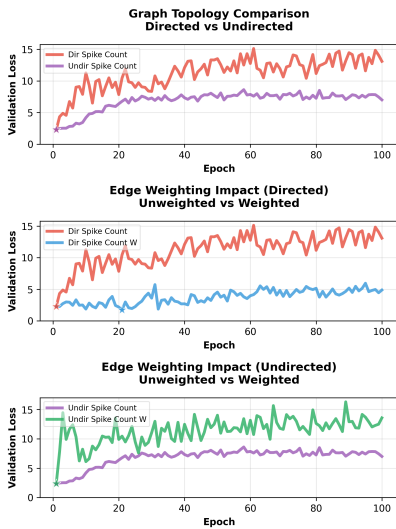


Fig. 8. Learning curves comparison for different graph construction methods at 50ms temporal resolution, demonstrating consistent performance trends across temporal windows.

Repository: https://github.com/Leeo1994/MSc_Dissertation.git

Contents:

- Graph Generator
- Tweaked model2 (tuner.py) and a few additional dependencies
- All graph topologies over different time bins for letter A
- raw data
- model ready data

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